

AI-POWERED

Wildfire Early Detection

Milestone 6

Group 8

Introduction

The AI-Powered Solution

Proactive Wildfire Early Detection

Our system continuously analyzes weather physics, landscape topography and vegetation dryness to calculate daily wildfire hazard levels across California.

Instead of overwhelming crews with alerts, it identifies the **Top 25 Highest-Risk Zones Every Day**. This provides organizations like CAL FIRE a crucial one-day head start to preposition resources and protect communities before ignition.

The Challenge

Why Early Detection Matters

01

The Speed of Wildfires

Under high winds and bone-dry air, a single spark can transform into a roaring blaze in minutes.

02

Limited Response Resources

Emergency services operate on finite budgets and cannot patrol every mile of the state simultaneously.

03

Late Incident Notice

Waiting for human calls or satellite smoke detection often means responding too late to prevent catastrophic loss.

Data Sources

DATA INGESTION

Google Earth Engine

We have extracted high-resolution environmental data directly from the Google Earth Engine platform, spanning multiple satellite constellations and climate reanalysis streams to power our predictive modeling.

- 01 Copernicus DEM**
Terrain Surface & Slope Acceleration - 30 m (Aggregated to 0.25°)
- 02 Sentinel 2**
Vegetation Health & Moisture Indices - 10–20 m / ~5-day Revisit
- 03 Sentinel 5P**
Atmospheric Composition - 3.5 km × 5.5 km
- 04 ERA5**
Atmospheric & Surface Weather - 0.25° Grid / Daily (5d lag)
- 05 FIRMS**
Ground-Truth Target Fire Labels - 1km / Daily

Data Range & Split

GEOGRAPHY & RESOLUTION

Spatial Parameters

California Focus

Analysis is restricted strictly to the California state boundary.

Grid Resolution

Using ERA5 atmospheric reanalysis grid cells at 0.25-degree resolution.

Training Zones

High and medium hazard fire zones selected specifically for training.

CHRONOLOGICAL SPLIT

Temporal Partitioning

2019 - 2022

Training Phase

Base historical data utilized to train the machine learning models. Training is restricted exclusively to May through November, capturing the high-activity months when most wildfires occur.

2023 / 2024

Validation & Calibration

Intermediate years reserved for validation (2023) and calibration (2024).

2025

Out-of-Sample Testing

Strictly separated test partition to evaluate model forecasting capability.

Unified Data Fusion for Predictive Wildfire Modeling

Our multi-modal pipeline integrates 63 raw features from 4 primary streams:

- **ERA5 Weather: 25 variables**
- **Copernicus DEM: 8 terrain features**
- **Sentinel-2 Optical: 21 spectral bands**
- **Sentinel-5P Gas: 10 atmospheric cols**



DATA CHALLENGE

- **Missing data**
- **Class Imbalance**
- **Resolution differences**

PIPELINE RESOLUTION & COVERAGE

Extensive panel coverage spanning California grid cells at approximately 25 km spatial resolution.

63

RAW FEATURES

4

DATA SOURCES

2,557

PANEL DAYS

672

GRID CELLS

Data Preprocessing & Cleaning Pipeline

RAW INPUT: 63 COLUMNS ACROSS 672 GRID CELLS

STEP 01

Data Cleaning

Ensures baseline data integrity by performing critical sanitation operations:

- Drops constant/redundant columns.
- Clips soil moisture values based on physical and atmospheric constraints.

STEP 02

KNN Imputation

Resolves observational gaps using Spatial-Temporal K-Nearest Neighbors:

- Addresses Sentinel-2 optical cloud cover and orbital gaps.
- Leverages spatial autocorrelation to reconstruct missing indices without zero/median bias. Backed by Literature: Liu et al., Remote Sensing 2020 ([MDPI Article 1884](#))

STEP 03

Risk Stratification

Focuses modeling efforts on highly vulnerable, active fire zones:

- Filters study domain down to High/Medium risk classes([data source](#)).
- Reduces grid footprint from 672 original cells to 437 .

Feature Engineering & Pruning Pipeline

FEATURE CONTRACT: 107 CANDIDATES → 86 PRODUCTION

STEP 04

Feature Engineering & Causality

Temporal causality contracts applied with minimum 2-day lags to prevent lookahead bias:

- FIRMS neighboring cells & historical cells
- ERA5 14-day & 30-day meteorological rollups
- Sentinel 2 - 5-day observation window

107

Candidate Features

2d

Temporal Lag Contract

STEP 05

TreeSHAP Feature Pruning

Identifies and eliminates lowest contribution variables to secure a highly refined:

20%

Features Pruned

86

Active Production

Features

Minimizes multicollinearity and resource usage while preserving predictive performance.

Why LightGBM as Champion Algorithm?

CORE TECHNICAL ADVANTAGES

- **Efficiency on Grids:** Faster training speed and lower memory footprint on high-dimensional multi-sensor grids.
- **Data Robustness:** Native handling of continuous satellite features and sparse missing values without complex imputation.

BACKED BY LITERATURE

MDPI Remote Sensing (2022) proved LightGBM outperforms Random Forest and Logistic Regression.

0.935

88.8%

ROC-AUC

Accuracy

"A Forest Fire Susceptibility Modeling Approach Based on Light Gradient Boosting Machine Algorithm"

[Read Published Research Paper →](#)

WHY OUR ROADMAP OUTPERFORMS THE LITERATURE

A. 50/50 Balance vs. Real-World Class Imbalance

The Paper: Trained on an artificial 50/50 split (inflates metrics).

Our Model: Entire state of CA under realistic <1.5% fire imbalance.

B. Static Susceptibility vs. Dynamic 24h Forecasting

The Paper: Static spatial likelihood mapping of a single regional park.

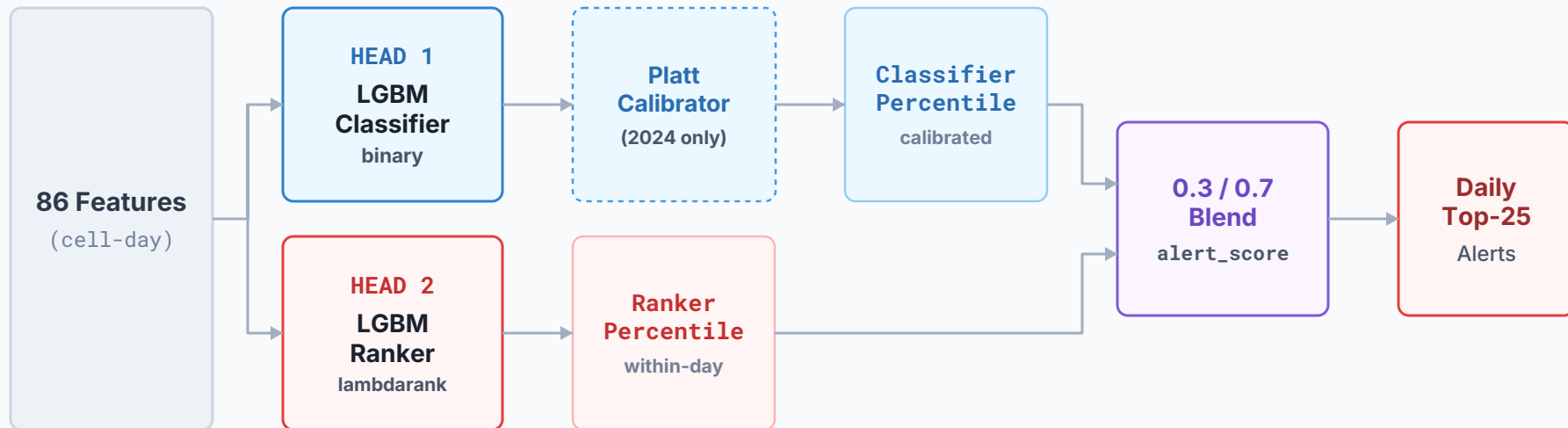
Our Model: Next-day dynamic temporal forecasting for proactive deployment.

C. Dual-Engine Architecture (Classifier + Ranker)

The Paper: Basic single-head standard classification.

Our Model: Dual-Head Classifier + LambdaRank with calibrated output.

Champion Model Pipeline & Alert Inference



Champion Model Performance

ROC-AUC

0.77

Separates fire vs no-fire well above chance

PR-AUC

0.145

Decent under rare events (~1.4% fire rate)

Recall@25

36%

Catches ~1 in 3 real fires if you alert top 25 cells/day

Precision@25

9%

~1 in 11 top-25 alerts is a true fire

False Alerts/Day

~23

Cost of that daily recall budget

Key Strategic Insight

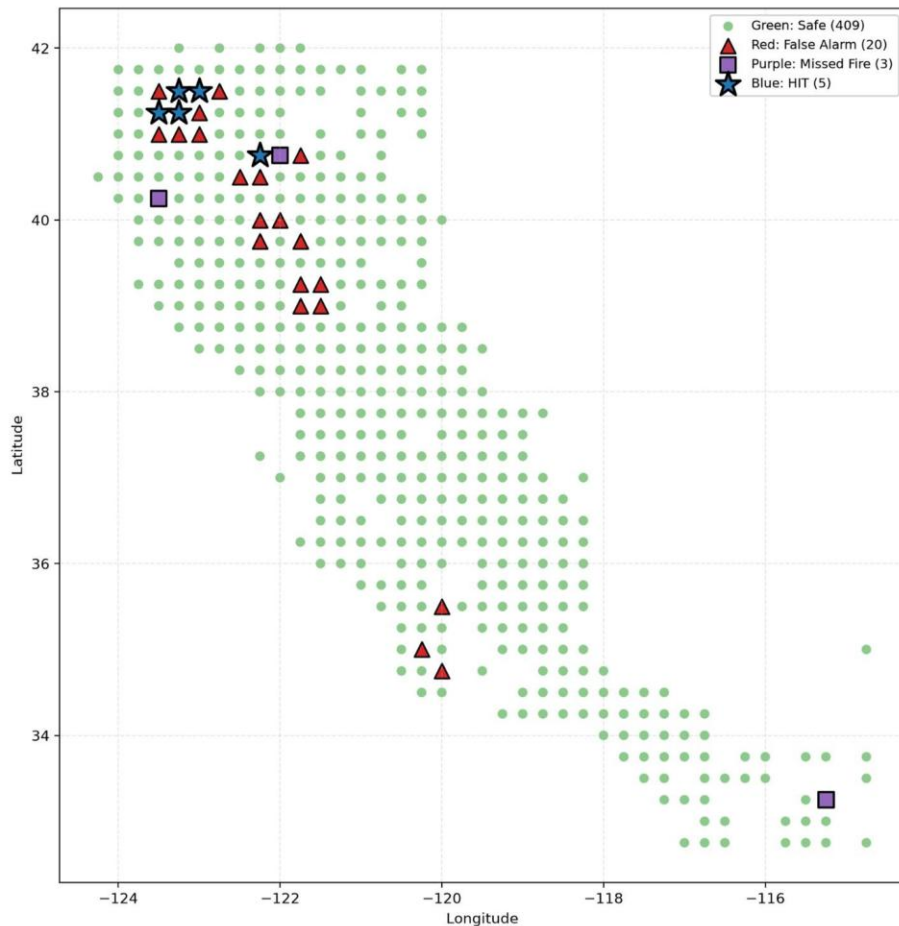
Optimal Ranking & Alerting Tool

This model is highly effective when used as a prioritized daily ranking system, but is not designed to act as a simple "fire/no-fire" binary switch.

LightGBM grid confusion map (437 cells)

Date: 2025-07-07

blend clf=0.30 / ranker=0.70



Champion Model Output

This is the output for July 7th 2025

- TP: 5
- TN: 409
- FP: 20
- FN: 3

This captured 5 out of 8 fires which is 62.5% of all fires on that day.

Interpreting Champion Results In Context

01 • LOW PROBABILITIES

Rarity Drives Low Scores

With a 1.42% prevalence, even our high-performing multimodal model (ERA5, S2/5P, DEM) correctly assigns near-zero risk to most grids.

02 • OBJECTIVE-DEPENDENT

Built for Top-K Ranking

A strict 0.5 cutoff is virtually useless here. The system is optimized as a Top-25 daily alerting tool such that recall is significantly improved.

03 • BASELINE COMPARISON

~15x Over Chance

Absolute scores look modest but our PR-AUC 0.145 vastly outperforms chance (0.009), persistence (0.041), and weather-only (0.096).

04 • TEMPORAL SHIFT

Unresolved Transferability

The 2023 to 2025 performance drop reflects honest future-held-out testing, revealing true real-world deployment challenges.

05 • GEOGRAPHIC ERROR

Physical, Non-Random Bias

Over-prediction concentrates in northern CA & misses follow Sierra terrain, showing error patterns are strictly shaped by seasonal regimes.

SUMMARY INSIGHT

Strong & Viable Tool

Our metrics define a highly useful daily ranking system that regionalizes risk, where moderate precision is the expected cost of catching rare fires.

Champion vs Other Models

MILESTONE 3

Multimodal Fusion

Accuracy: 81.50%

PR-AUC: 0.531

ROC-AUC: 0.831

Recall: 76.20%

Evaluated on 4:1 sampled sub-dataset; S2 CNN + S5P CNN + ERA5/DEM LSTM

MILESTONE 4

V4 LightGBM

Accuracy: 98.24%

PR-AUC: 0.1751

ROC-AUC: 0.842

Recall@25: 38.33%

Full grid evaluation; optimized directly for daily alert budget (Recall@25) - Test for 2025

MILESTONE 5

LightGBM Champion

Accuracy: 98.54%

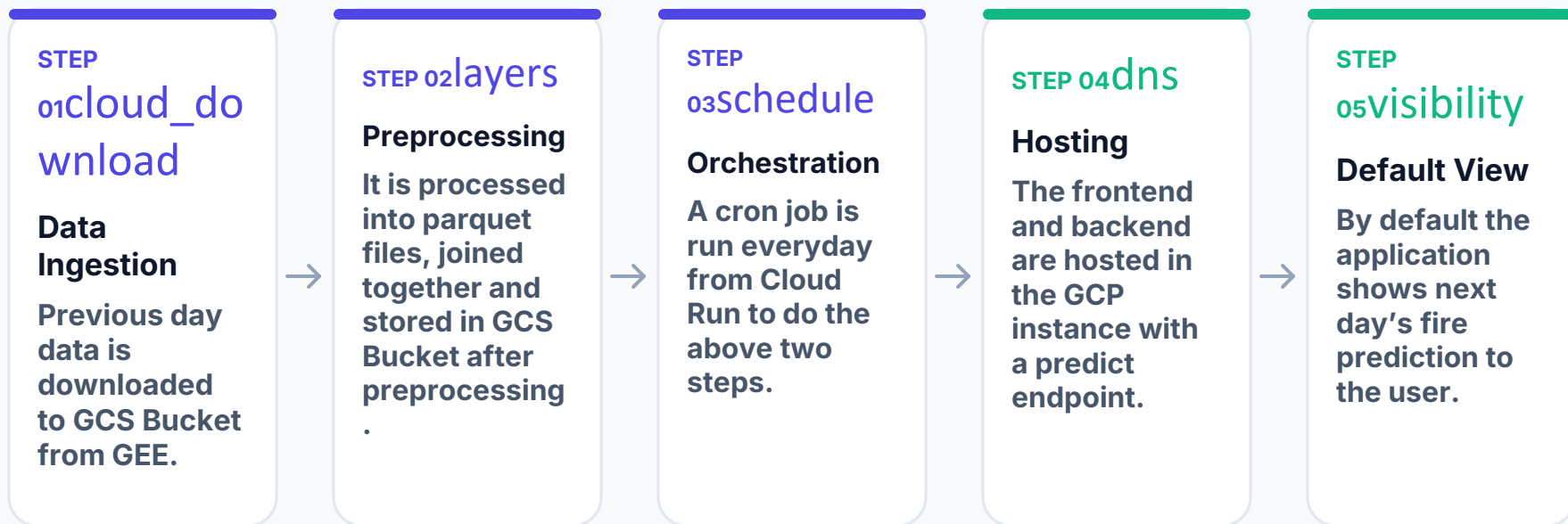
PR-AUC: 0.1451

ROC-AUC: 0.7718

Recall@25: 36.38%

Stage C KNN pre-imputation, May-Nov fire season, high/med fire cells - Test for 2025

Deployment



DEMO

References

- [A Forest Fire Susceptibility Modeling Approach Based on Light Gradient Boosting Machine Algorithm](#)
- [A Comprehensive Survey of Satellite-Based Wildfire Indicators and Spatiotemporal Modeling Approaches: Past, Present, and Future](#)
- [EGUsphere - Direct Estimation of Wildfire Emissions at High Latitudes from Combined Polar Orbiter FRP and Sentinel-5P CO Data](#)

Signatures

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